

MASTER OF ARTS (PSYCHOLOGY)**Term-End Examination, 2019****MPC-006 : STATISTICS IN PSYCHOLOGY****Time : 2 Hours]****[Maximum Marks : 50**

Note : All sections are compulsory. Use of simple calculator is permitted.

SECTION-A

Note : Answer any two of the following questions in about 450 words each.

1. Explain descriptive statistics. Describe graphical presentation of data. [3+7=10]
2. Discuss Spearman's rank-order correlation with suitable example. [10]
3. Describe standard error. Compute t test for the following data : [3+7=10]

Data 1 :	8,	4,	2,	1,	6,	7,	2,	3,	2,	5
Data 2 :	10,	15,	20,	21,	9,	30,	31,	19,	10,	15

Critical value = 2.10 at 0.05 level 2.88 at 0.01 level

4. What is tetrachoric correlation ? Compute Kendall's tau for the following data : [3+7=10]

	A	B	C	D	E
Variable X:	3	1	2	4	5
Variable Y:	4	5	3	2	1

SECTION-B

Note : Answer **any four** of the following questions in about **250** words each.

5. Explain hypothesis testing. Describe errors in hypothesis testing. [3+3=6]
6. Compute Mann-Whitney U test for the following data : [6]

Group 1 : 32, 93, 19, 40, 46, 71, 63, 98, 47, 49, 52

Group 2 : 89, 21, 82, 91, 85, 84, 103, 101, 100, 107, 110

7. Compute Chi-square for the following data : [6]

Opinion			
	Agree	Undecided	Disagree
Male	10	13	11
Female	15	12	14

8. Discuss the use of non-parametric tests. [6]
9. Explain in detail Wilcoxon matched pair signed rank test with suitable example. [6]

SECTION-C

Note : Write short notes on **any two** of the following in about **100** words each.

10. Semipartial Correlation. [3]
11. Factors causing divergence in normal curve. [3]
12. Meaning of Variance. [3]

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MASTER OF ARTS (PSYCHOLOGY)**Term-End Examination, 2019****MPC-006 : STATISTICS IN PSYCHOLOGY****Time : 2 Hours****Maximum Marks : 50**

Note : All sections are compulsory. Use of simple calculator be permitted.

SECTION - A

Note : Answer any two of the following in about 450 words each:

[2×10=20]

1. Explain the meaning of inferential statistics. Describe hypothesis testing. [5+5]
2. Discuss in detail partial correlation with suitable example. [10]
3. Describe the basic assumptions in testing of significance of difference between two sample means. Compute t-value for the following data : [4+6]



Data 1:	2	3	4	2	5	3	4	2	2	3
Data 2:	3	4	5	5	5	2	2	4	4	6

Critical Value =

2.10 at 0.05 level of significance

2.88 at 0.01 level of significance.

4. Compare between Rho and Tau. Compute Kendall's tau for the following data : [3+7]

	A	B	C	D	E
Variable X :	3	2	4	1	5
Variable Y :	5	1	2	3	4

SECTION - B

Note : Answer **any four** of the following questions in about **250** words each : [4×6=24]

5. Explain the meaning and concept of levels of significance. Describe the steps in setting up the level of significance [3+3]
6. Compute Mann Whitney U test for the following data : [6]

Group 1 :	19	25	17	27	30	16	44	50	
Group 2 :	12	36	10	37	13	46	57	63	70

7. Compute chi-square for the following data : [6]

	Agree	Disagree
Males	12	14
Females	13	20

8. Define non-parametric statistics and discuss its assumptions. [6]
9. Explain divergence in normality with the help of suitable diagrams. [6]

SECTION-C

Note : Write short notes on **any two** of the following in about 100 words each : [2×3=6]

10. Measures of Dispersion [3]
11. Concept of Normal Curve [3]
12. Kruskal-Wallis ANOVA test [3]

-----X-----

MASTER OF ARTS (PSYCHOLOGY)

Term-End Examination

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours]

[Maximum Marks : 50

Note: All Sections are compulsory. Use of simple calculator is permitted.

Section-A

Answer any two of the following questions in about 450 words each: 2×10=20

1. Define parametric statistics and describe its assumptions, advantages and disadvantages. 3+7
2. Explain linear and nonlinear relationship with suitable diagrams. Discuss the steps in computing Pearson's product moment correlation. 4+6
3. Compute *t* test for the following data: 10
Group-A 2, 3, 5, 4, 1, 5, 10, 4, 6, 10
Group-B 7, 10, 5, 8, 4, 6, 12, 13, 2, 3
4. Compute one way ANOVA for the following data and indicate if groups differ on the variable: 10

$$F \text{ value} = \begin{array}{cc} P < 0.5 & P < 0.1 = \\ 3.35 & 5.49 \end{array}$$

Group A: 2, 3, 4, 2, 3, 2, 2, 2, 3, 3

Group B : 2, 4, 5, 5, 5, 2, 3, 5, 5, 2

Group C : 2, 3, 4, 2, 5, 2, 2, 3, 2, 3

Section-B

Answer any four of the following questions in about 250 words each. $4 \times 6 = 24$

5. Discuss frequency distribution in terms of grouped and ungrouped data. Elucidate the types of frequency distribution. $4+2$
6. Differentiate between partial and part correlation with suitable example. 6
7. Compute Mann Whitney U for the following data:
Group 1 : 23, 21, 7, 14, 10, 13, 25, 29, 48, 55
Group 2 : 20, 8, 15, 9, 45, 12, 40, 47, 50, 51
8. Discuss the step by step procedure for Kendall's Rank Order Correlation. 6
9. Compute Chisquare for the following data:

	Male	Female
Literate	10	30
Illiterate	20	40

Section-C

Write short notes on any two of the following in about 100 words each: $2 \times 3 = 6$

10. Sampling error. 3
11. Assumptions underlying the Analysis of Variance. 3
12. Nominal data. 3

No. of Printed Pages : 3

MPC-006

M. A. (PSYCHOLOGY)

(MAPC)

Term-End Examination

June, 2021

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : (i) All Sections are compulsory.

(ii) Use of simple calculator is permitted.

Section—A

Note : Answer any **two** of the following questions in about **450** words each. $2 \times 10 = 20$

1. Describe the **four** major statistical techniques for organising the data. 10
2. Discuss the concept, properties and characteristics of normal probability curve with the help of a suitable diagram. 10

P. T. O.

[2]

MPC-006

3. Explain Phi coefficient. Compute Pearson's product moment correlation for the following data : 2+8

Individuals	A	B	C	D	E	F	G	H	I	J
Data X	5	5	4	4	4	3	2	2	6	5
Data Y	6	6	4	4	4	5	5	5	4	7

4. Compute one-way ANOVA for the following data : 10

Group A	4	5	5	5	3	3	3	4	4	4
Group B	5	5	6	5	6	3	4	7	7	7
Group C	6	5	6	3	8	8	7	5	3	9

Critical value :

3.35 at 0.05 level

5.49 at 0.01 level.

Section—B

Note : Answer any **four** of the following in about **250** words each. $4 \times 6 = 24$

5. Differentiate between parametric and non-parametric statistics. 6
6. Compute Mann-Whitney U-test for the following data : 6

Group A : 60, 22, 27, 31, 35, 47, 56, 40, 55

Group B : 20, 29, 24, 39, 34, 41, 49, 50, 51

[3]

7. Compute Chi-square for the following data : 6

Responses		
	Yes	No
Male	20	15
Female	20	20

8. Elucidate interactional effect. Discuss the merits and demerits of two-way ANOVA. 2+4
9. Describe the levels of measurement with suitable examples. 6

Section—C

Note : Write short notes on any *two* of the following in about **100** words each. 2×3=6

10. Decision errors. 3
11. Characteristics of variance. 3
12. Multiple regression. 3

No. of Printed Pages : 5

MPC-006

M. A. (PSYCHOLOGY)
(MAPC)

Term-End Examination

December, 2021

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *All Sections are compulsory.*

(ii) *Use of simple calculator is permitted.*

Section—A

Note : *Answer any **two** of the following questions in about **450** words each.*

2×10=20

1. Discuss the applications of normal distribution curve. Describe divergence from normality with

suitable diagram and discuss the factors causing divergence. 2+8

2. Elucidate the definition and assumptions of non-parametric statistics and describe its use. 7+3

3. Explain biserial correlation. Compute Pearson's product moment correlation for the following data : 3+7

	Data X	Data Y
A	10	12
B	12	13
C	13	5
D	11	12
E	9	13
F	8	15
G	12	10
H	5	10
I	10	10
J	10	10

[3]

MPC-006

4. Compute one-way ANOVA for the following data : 10

Group 1	Group 2	Group 3
2	4	5
3	5	5
3	5	5
2	5	4
2	2	4
3	4	5
3	4	5
2	5	4
2	5	4
2	5	5

CV = 3.35 at 0.05 level

5.49 at 0.01 level.

Section—B

Note : Answer any **four** of the following in about **250 words each**. 4×6=24

5. Compute Mann-Whitney U-test for the following data :

P. T. O.

[4]

MPC-006

Data A	Data B
5	4
23	17
20	18
11	8
27	29
24	13
40	30
37	33
45	50
31	

6. Compute Chi-square for the following data : 6

Responses

	Yes	No	Undecided
Male	10	10	10
Female	20	10	10

7. Define and describe coefficient of correlation. Discuss the characteristics and measures of correlation. 3+3

8. Describe the measures of central tendency and measures of dispersion. 3+3
9. Explain step by step procedure for computation of Kruskal Wallis ANOVA with an example. 6

Section—C

Note : Write short notes on any **four** of the following in about **100** words each. $2 \times 3 = 6$

10. Degrees of Freedom 3
11. Interactional Effect 3
12. The Regression Equation 3

MASTER OF ARTS (PSYCHOLOGY)

Term-End Examination

June, 2022

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 hours

Maximum Marks : 50

Note : *All sections are **compulsory**. Use of simple calculator is permitted.*

SECTION A

*Answer any **two** of the following questions in about 450 words each :* *2×10=20*

1. Explain non-parametric statistics with a focus on its assumptions, advantages and disadvantages. 10

2. Discuss the steps in computation of regression with the help of suitable example. 10

3. Compute Kendall's tau for the following data : 10

Data X: 7, 8, 5, 6, 3

Data Y: 4, 6, 3, 1, 2

4. Compute ANOVA for the following data : 10

Group 1 : 2, 3, 4, 5, 2, 3, 4, 3, 2, 1

Group 2 : 3, 4, 3, 2, 6, 5, 4, 3, 3, 2

Group 3 : 4, 5, 4, 3, 3, 3, 3, 3, 3, 3

SECTION B

Answer any **four** of the following questions in about 250 words each :

4×6=24

5. Describe biserial and tetrachoric correlation. 3+3
6. Discuss the meaning, importance and application of standard error. 6
7. Compute Chi-square for the following data : 6

Responses

	Yes	No
Males	4	6
Females	6	4

8. Discuss Kruskal-Wallis analysis of variance. 6
9. Describe the importance and application of normal distribution. 6

SECTION C

*Write short notes on any **two** of the following in about 100 words each :*

2×3=6

- | | |
|---|----------|
| 10. Types of Frequency Distribution | 3 |
| 11. Linear and Non-linear Relationship | 3 |
| 12. Kurtosis | 3 |
-

MASTER OF ARTS (PSYCHOLOGY)**Term-End Examination****December, 2022****MPC-006 : STATISTICS IN PSYCHOLOGY***Time : 2 hours**Maximum Marks : 50*

Note : *All sections are **compulsory**. Use of simple calculator is permitted.*

SECTION A

*Answer any **two** of the following questions in about 450 words each :*

2×10=20

1. Describe the measures of central tendency and measures of dispersion. *5+5*
2. Explain the steps in the calculation of Pearson's product-moment correlation, with the help of an example. *10*
3. Compute Kendall's 'tau' for the following data : *10*

	A	B	C	D	E
Data X	5	3	7	6	2
Data Y	7	8	9	6	4

4. Compute ANOVA for the following data :

10

Group 1 : 4, 2, 4, 5, 4, 3, 4, 4, 2, 1

Group 2 : 3, 4, 4, 2, 6, 3, 4, 4, 3, 2

Group 3 : 1, 5, 4, 3, 2, 2, 1, 1, 3, 3

SECTION B

Answer any **four** of the following questions in about 250 words each :

4×6=24

5. Discuss the characteristics of normal probability curve. 6
6. Explain Type I and Type II errors, with suitable examples. 6
7. Compute chi-square for the following data : 6

		Responses	
		Yes	No
Gender	Males	5	5
	Females	6	4

8. Describe point-biserial correlation and phi-coefficient. 6
9. What is Skewness ? Explain the factors causing divergence in normal distribution. 2+4

SECTION C

Write short notes on any **two** of the following in about 100 words each :

2×3=6

- | | |
|-----------------------------------|---|
| 10. Point and Interval Estimation | 3 |
| 11. Partial Correlation | 3 |
| 12. Mann-Whitney U-test | 3 |
-

No. of Printed Pages : 4

MPC-006

M. A. (PSYCHOLOGY)
(MAPC)

Term-End Examination

June, 2023

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *All Sections are compulsory.*

(ii) *Use of simple calculator is permitted.*

Section—A

Note : *Answer any **two** of the following questions in about **450** words each (wherever applicable).*

2×10=20

1. Elucidate parametric and non-parametric statistics. Describe the assumptions of parametric statistics. 10
2. Explain point biserial correlation and phi-coefficient with suitable examples. 5+5

P. T. O.

3. Compute one way ANOVA for the following data : 10

Group A	Group B	Group C
4	6	10
6	7	11
7	10	12
9	12	13
2	13	14
4	14	15
7	15	17
8	17	10
9	18	2
2	3	4

Critical value for $F = 3.35$ at 0.05 level of significance.

Critical value for $F = 5.49$ at 0.01 level of significance.

4. Compute Spearman's rho for the following data :

10

Data A	Data B
16	15
19	17
18	16
10	9
12	10
13	12
17	19
9	8
7	6
5	4

Section—B

Note : Answer any **four** question in about **250** words each (wherever applicable).

4×6=24

5. Explain the characteristics of normal probability curve. 6
6. Compute Chi-square for the following data : 6

Responses

Gender	Yes	No
Male	15	5
Female	10	10

7. Discuss the importance and application of standard error of means. 6
8. Compute Mann-Whitney U test for the following data : 6

Group A	Group B
21	12
30	16
26	17
32	19
41	20
35	24
15	31

9. Explain the concept of interactional effect. Discuss merits and demerits of two way ANOVA. 2+6

Section—C

Note : Answer any **two** of the following in about 100 words each. $2 \times 3 = 6$

10. Hypothesis testing. 3
11. Outliers and curvilinearity. 3
12. Wilcoxon matched pair signed rank test. 3

No. of Printed Pages : 4

MPC-006

M. A. (PSYCHOLOGY)
(MAPC)

Term-End Examination

December, 2023

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *All Sections are compulsory.*

(ii) *Use of simple calculator is permitted.*

Section—A

Note : *Answer any **two** of the following questions in about **450** words each (wherever applicable).*

2×10=20

1. Explain organisation of data. 10
2. Elucidate partial and part correlation (semi-partial correlation) with the help of suitable examples. 5+5

P. T. O.

3. Compute one-way ANOVA for the following data : 10

Group A	Group B	Group C
7	2	3
8	4	4
6	7	2
2	8	2
4	6	3
5	4	4
7	7	4
8	8	5
9	9	7
10	10	2

Critical values :

3.35 at 0.05 level of significance

5.49 at 0.01 level of significance

4. Compute Spearman's rho for the following data : 10

Data A	Data B
15	20
16	17
11	4
13	12
18	5
10	11
12	10
6	8
7	2
2	3

Section—B

Note : Answer any **four** questions in about **250** words each (wherever applicable).

$$4 \times 6 = 24$$

5. Describe divergence in normality with the help of suitable diagrams. 6
6. Compute Chi-square for the following data : 6

Responses

	Agree	Undecided	Disagree
Junior Managers	10	15	5
Senior Managers	10	5	15

7. Describe the levels of measurement with suitable examples. 6
8. Compute Mann-Whitney U-test for the following data : 6

Group A	Group B
20	31
13	10
14	9
16	7
18	6
21	3
30	25

9. Elucidate hypothesis testing. 6

Section—C

Note : Write short notes on any **two** of the following
in about **100** words each. $2 \times 3 = 6$

10. Tetrachoric correlation. 3
11. Linear regression 3
12. Scatter plot 3

No. of Printed Pages : 3

MPC-006

M. A. PSYCHOLOGY (MAPC)

Term-End Examination

June, 2024

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : *All Sections are compulsory. Use of simple calculator is permitted.*

Section—A

Note : Answer any **two** of the following in about
450 words each : 2×10=20

1. Discuss the assumptions of parametric and nonparametric statistics. 10
2. Explain the concept of normal curve with the help of a diagram. Describe the characteristics of normal probability curve. 3+7

P. T. O.

3. Compute ANOVA for the following data : 10

Group A	6	7	3	4	2	4	6	7	8	1
Group B	4	7	3	4	7	3	2	7	4	2
Group C	3	6	2	6	8	6	3	9	6	3

Critical value : 3.35 at 0.05 level

5.49 at 0.01 level

4. Compute Chi-square for the following data : 10

Responses				
Categories	Never	Rarely	Sometimes	Always
Group A	10	12	8	10
Group B	6	4	7	3
Group C	11	9	4	6
Group D	7	3	4	6

Section—B

Note : Answer any **four** of the following in about
250 words each : $4 \times 6 = 24$

5. Compute Spearman's rho for the following data : 6

Data A	16	17	4	6	7	5	10	2	3
Data B	18	19	21	4	7	8	9	6	2

[3]

6. Compute Mann-Whitney U for the following data : 6

Group A	7	6	3	2	14	6	10
Group B	7	8	1	13	4	7	9

7. Elucidate part correlation. 6
8. Explain the concept, merits and demerits of two- way ANOVA. 6
9. Explain the measures of central tendency and measures of dispersion. 3+3

Section—C

Note : Write short notes on any **two** of the following in about **100** words each : 2×3=6

10. Level of significance 3
11. Kruskal-Wallis ANOVA test 3
12. Standard error 3

No. of Printed Pages : 4

MPC–006

M. A. (PSYCHOLOGY)
(MAPC)

Term-End Examination

December, 2024

MPC–006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *All Sections are compulsory.*

(ii) *Use of simple calculator is permitted.*

Section—A

Note : Answer any **two** of the following questions in
about **450** words each. 2×10=20

1. Explain parametric and non-parametric statistics. Describe the advantages and disadvantages of non-parametric statistics. 5+5
2. Explain the organisation of data with the help of suitable examples. 10

3. Compute ANOVA for the following data : 10

Group A	Group B	Group C
6	4	6
2	3	7
3	2	8
4	1	1
1	3	3
3	4	2
2	3	4
3	2	6
2	1	7

Critical value :

3.35 at 0.05 level

5.49 at 0.01 level.

4. Compute Chi-square for the following data : 10

Responses

Categories	Agree	Undecided	Disagree
Group A	8	2	5
Group B	6	4	5
Group C	2	3	5
Group D	5	5	5

Section—B

Note : Answer any **four** of the following questions
in about **250** words each. 4×6=24

5. Compute Spearman's rho for the following data : 6

Data A	Data B
1	3
7	9
6	12
4	11
5	8
9	13
8	15
2	14
3	4
10	2

6. Explain the concept of normal distribution. Describe the factors causing divergence in normality. 3+3
7. Describe point biserial correlation with the help of suitable examples. 6

8. Describe Wilcoxon matched pair signed rank test. 6
9. Describe interactional effect. Explain the demerits of two-way ANOVA. 2+4

Section—C

Note : Write short notes on any *two* of the following in about **100** words each. $2 \times 3 = 6$

10. Type I and Type II errors. 3
11. Multiple regression. 3
12. Mann-Whitney U-test. 3

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No. of Printed Pages : 5

MPC-006

M. A. (PSYCHOLOGY)

(MAPC)

Term-End Examination

June, 2025

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : *All Sections are compulsory. Use of simple calculator is permitted.*

Section—A

Note : *Answer any **two** of the following questions in about **450** words each (wherever applicable).* **2×10=20**

1. Explain classification and organisation of data with suitable examples. **10**

2. Compute one-way ANOVA (parametric test)
for the following data : 10

Group A	Group B	Group C
2	3	4
2	3	4
3	4	3
3	3	3
2	2	2
2	2	2
2	2	2

3. Compute Spearman's rho for the following
data : 10

Data A	Data B
3	2
4	8
6	7
5	3
8	12

13	4
9	15
1	1
12	10
11	9

4. Describe the assumptions of parametric and non-parametric statistics. 10

Section—B

Note : Answer any **four** of the following questions in about **250** words each (wherever applicable). 4×6=24

5. Compute Mann-Whitney U-test for the following data : 6

Group A	Group B
23	8
25	24
28	26
30	27

11	36
10	31
9	1
4	13
7	5
6	12

6. Compute Chi-square for the following data :

6

Gender	Responses	
	Yes	No
Males	10	5
Females	15	20

7. Discuss the errors in hypothesis testing. 6
8. Explain the characteristics of normal probability curve. 6
9. Describe the merits and demerits of two-way ANOVA. 3+3

Section—C

Note : Write short notes on any **two** of the following in about **50** words each. $2 \times 3 = 6$

- | | |
|---------------------------|---|
| 10. Level of significance | 3 |
| 11. Kurtosis | 3 |
| 12. Kendall's tau | 3 |

× × × × ×